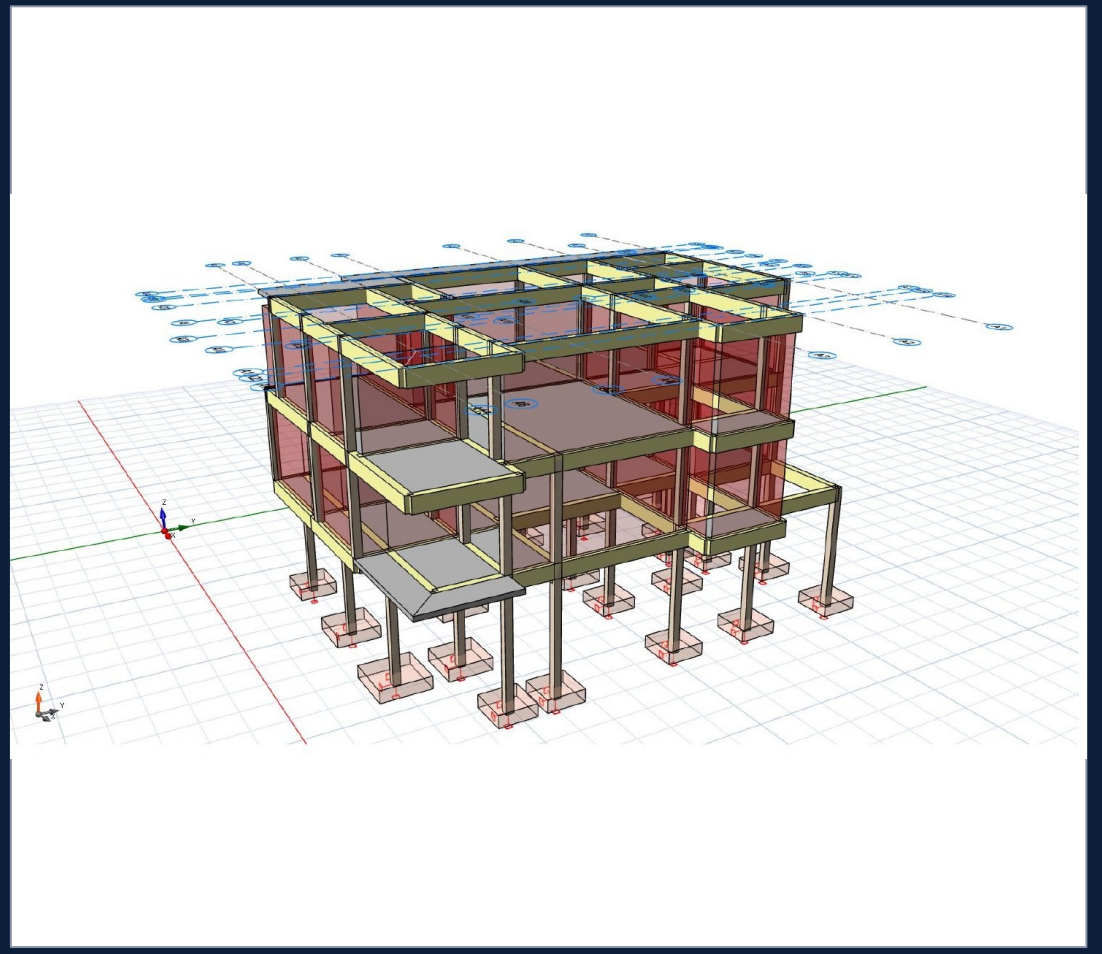
STRUCTURAL ENGINEERING PORTFOLIO

# Fahdil Dagold

## STRUCTURAL DESIGN ENGINEER

Structural modelling, analysis, reinforced concrete design and detailing



ProtaStructure 3D model of the triplex residential building

*Featured project: triplex residential building*

# Engineering profile

Structural Design Engineer with practical experience in structural modelling, analysis, reinforced concrete design and detailing. Work covers beams, columns, slabs, pad foundations and staircases, from interpreting architectural drawings through ProtaStructure modelling and design to AutoCAD reinforcement drawings.

*This portfolio presents one project in full: a triplex residential building. Each part of the work is marked as performed or reviewed.*

**FOCUS**
Reinforced concrete buildings

**DESIGN CODE**
BS 8110-1997

**SOFTWARE**
ProtaStructure 2021, AutoCAD 2024

**EDUCATION**
B.Tech Civil Engineering, LAUTECH, 2024

# Technical competencies

## Modelling and analysis

- ProtaStructure modelling and structural analysis
- 3D structural and analytical models
- Load cases, combinations and design checks

## Reinforced concrete design

- Slab design and reinforcement
- Beam and column design
- Pad footing and foundation design

## Detailing and drawings

- AutoCAD structural detailing
- Architectural drawing interpretation
- Structural documentation

## Review and site

- Exposure to design reviews
- Exposure to site supervision

SOFTWARE

**ProtaStructure 2021**
Modelling, analysis, design

**AutoCAD 2024**
Structural detailing and drawings

# Contribution on the triplex project

**PERFORMED** Work I carried out myself.

**REVIEWED** Work checked as part of the engineering process.

| Area of work | Scope | Contribution |
|---|---|---|
| Slab design and reinforcement | First and second floor slabs: design, reinforcement and layouts | PERFORMED |
| Structural modelling and analysis | ProtaStructure model, loading, load combinations, analysis | PERFORMED |
| Beam and column design | Design and reinforcement of frame members | PERFORMED |
| Foundation design | Pad footings, foundation layout and sections | PERFORMED |
| AutoCAD detailing | Foundation, framing, beam, column and staircase sheets | PERFORMED |
| Issued drawing set | 22 sheets, checked before issue | REVIEWED |

# Project overview

| Item | Detail |
|---|---|
| Building type | Triplex residential building |
| Structural system | Reinforced concrete frame on pad footings |
| Levels | Foundation, ground slab, first floor, second floor, roof |
| Columns | 230 x 230 mm |
| Beams | 230 x 450 mm and 230 x 600 mm |
| Slabs | 150 mm thick |
| Design codes | BS 8110-1997 (concrete), BS 6399 (loading) |
| Assumed allowable soil pressure | 200 kN/m2 |
| Analysis and design | ProtaStructure 2021 |
| Detailing | AutoCAD 2024, 22 sheets |

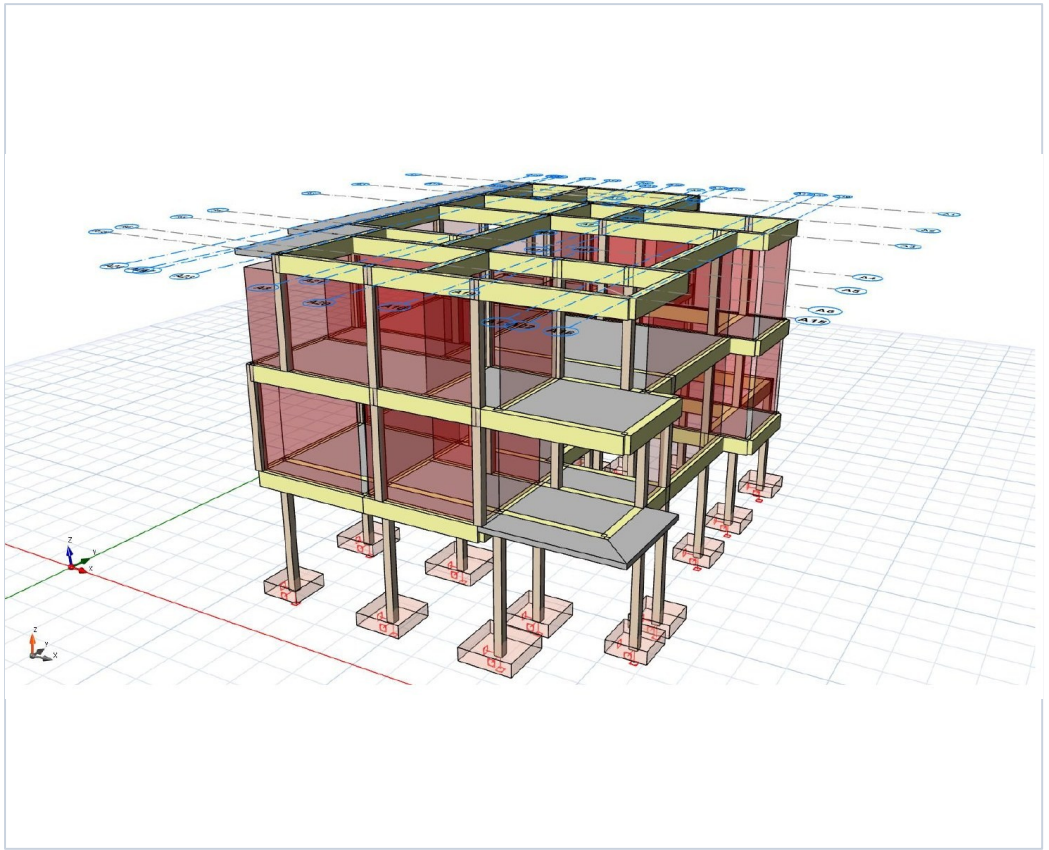


Figure 1. ProtaStructure 3D model: pad footings, 230 x 230 mm columns, beams, slabs and wall panels.

# Design approach and workflow

1

## Architectural input

Plans, sections and elevations interpreted to set the grid, column positions and beam layout.

2

## Structural model

Grids, storeys, columns, beams, slabs, wall panels and footings built in ProtaStructure.

3

## Loading and analysis

Load cases and combinations applied to the analytical model. Codes: BS 6399 and BS 8110-1997.

4

## Design

Slabs, beams, columns and pad footings designed to BS 8110-1997.

5

## Detailing

Reinforcement details drawn in AutoCAD for every structural element.

6

## Documentation

22 drawing sheets covering foundation, framing, slabs, beams, columns and stair.

PERFORMED

# Structural modelling: storey plans

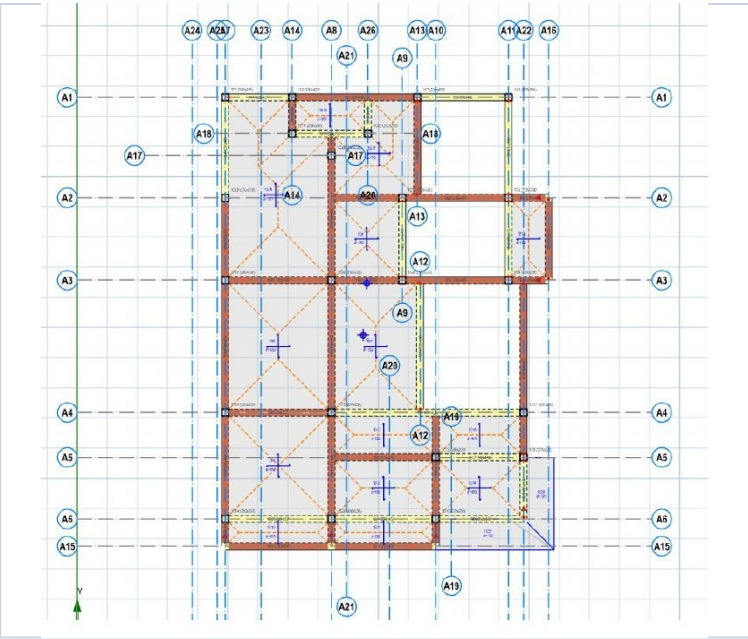


Figure 2. First floor plan

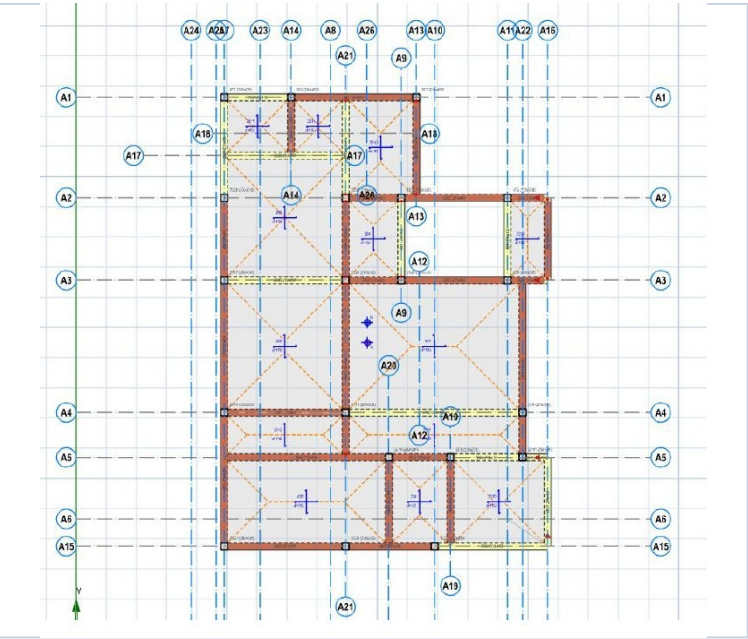


Figure 3. Second floor plan

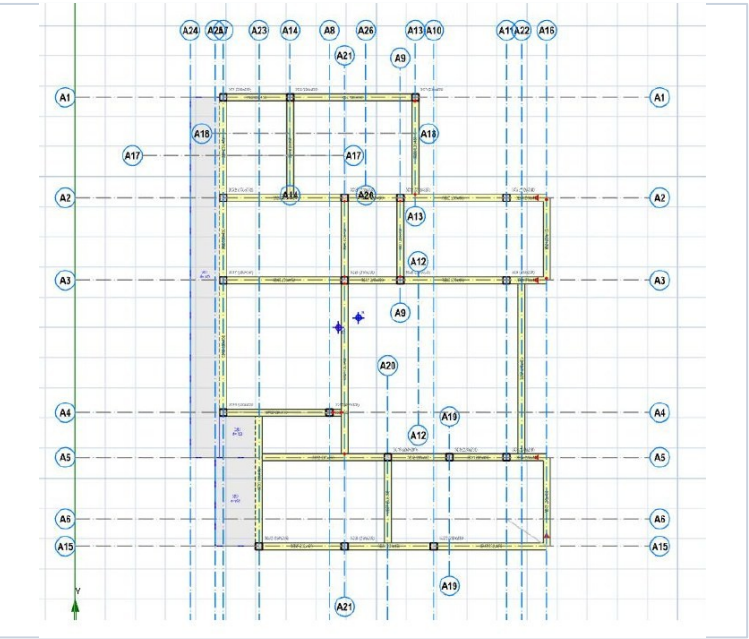


Figure 4. Roof plan

- Column grid, beams and slab panels set up storey by storey from the architectural layout.
- Dashed diagonals are the load distribution lines of each slab panel.
- Section sizes follow on the next page.

PERFORMED

# Column, beam and slab modelling

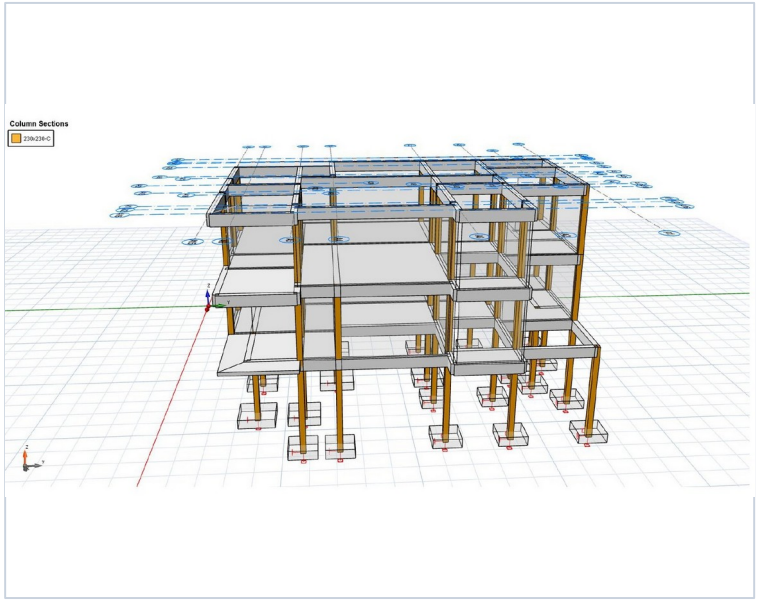



Figure 5. Column sections: 230 x 230 mm

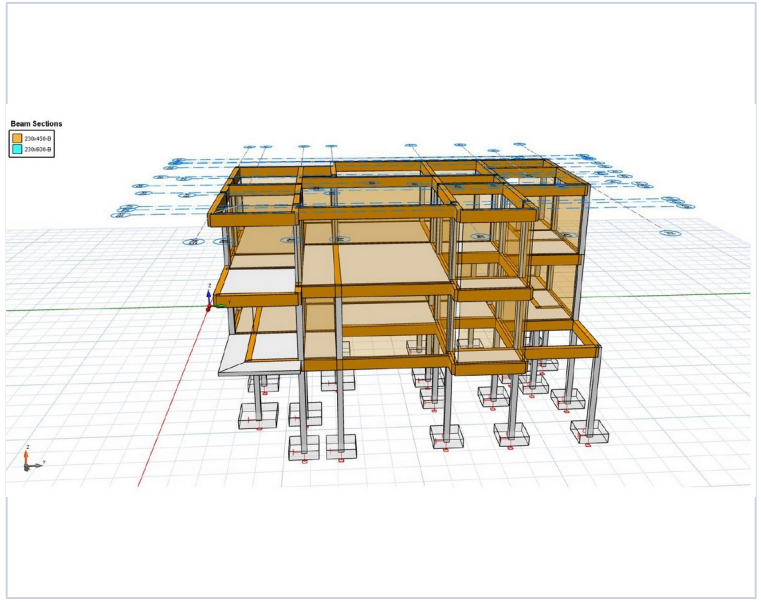



Figure 6. Beam sections: 230 x 450 mm and 230 x 600 mm

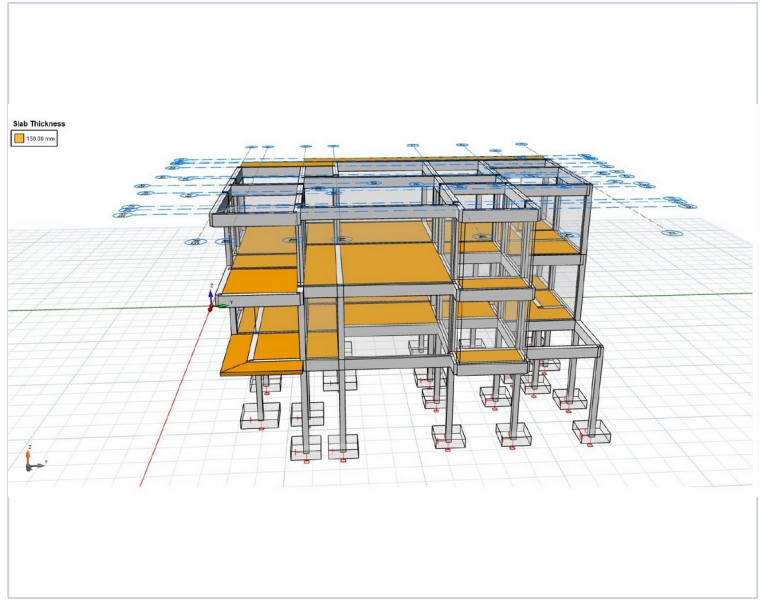



Figure 7. Slab thickness: 150 mm

- One column section is used throughout: 230 x 230 mm.
- Beams are 230 x 450 mm, with 230 x 600 mm on selected members.
- Slabs are modelled at 150 mm and span between the supporting beams.

PERFORMED

# Three-dimensional structural model

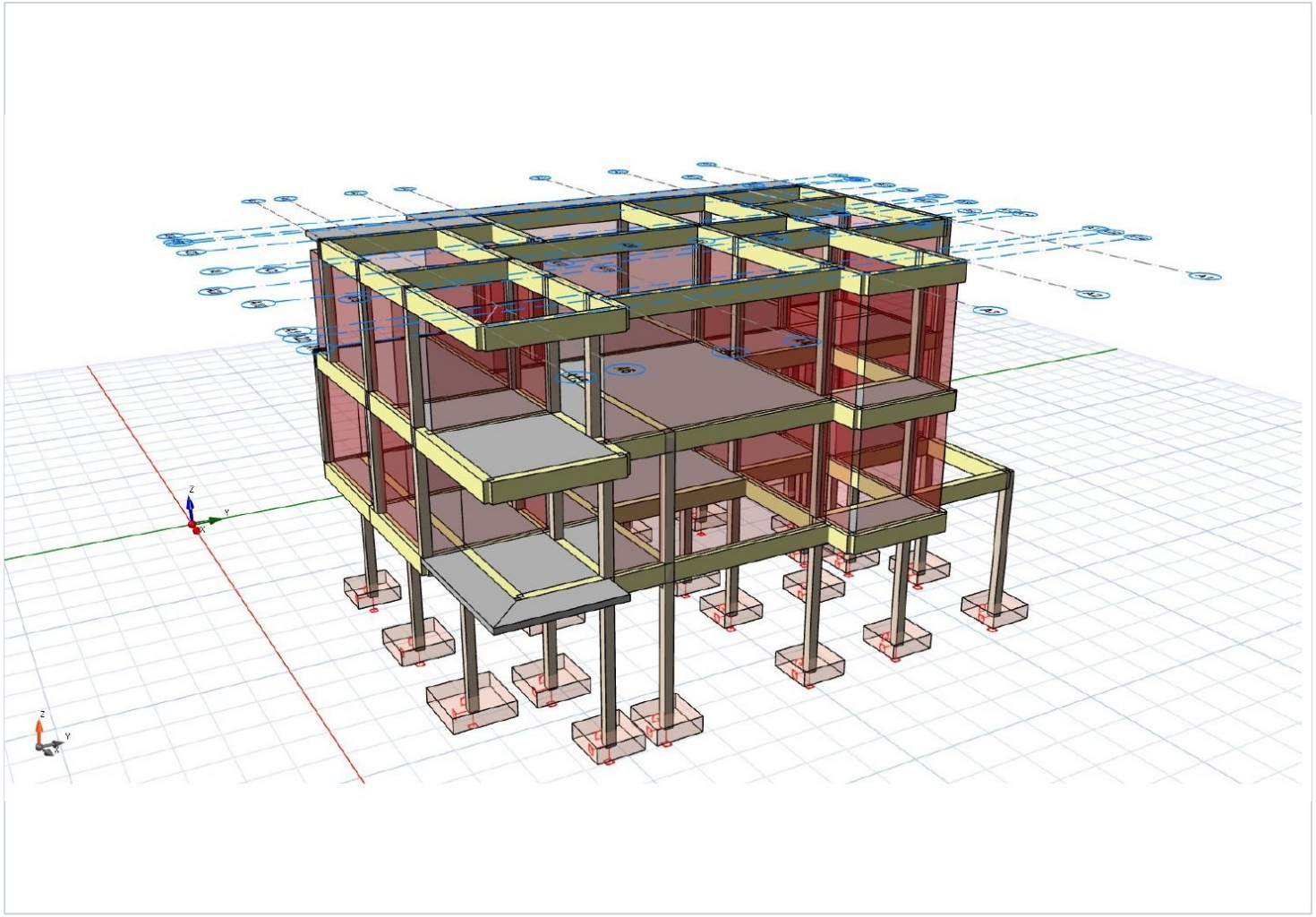


Figure 8. 3D model: pad footings, columns, beams, slabs and wall panels (translucent red).

- Pad footings sit under every column.
- Wall panels are modelled between beams.
- Slab panels span between the beams on each level.
- The analytical model and design modules are generated from this model.

PERFORMED

# Analytical model, loading and load combinations

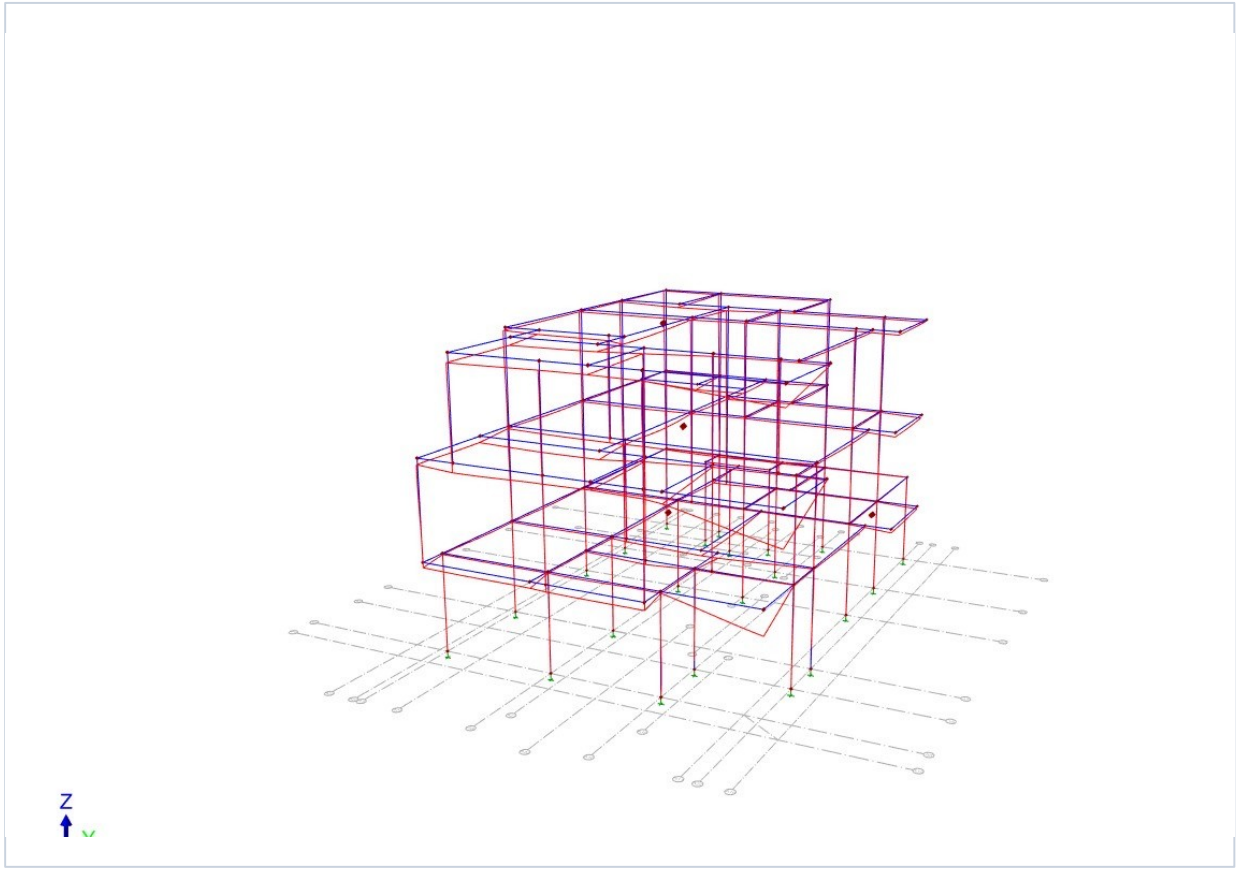



Figure 9. Analytical (frame) model generated by ProtaStructure: columns, beams and slab members, with supports at the column bases.

| Combination | Factors |
|---|---|
| G+Q | 1.4G + 1.6Q |
| (G+Q)p1 | 1G + 0.4Gp1 + 1.6Qp1 |
| (G+Q)p2 | 1G + 0.4Gp2 + 1.6Qp2 |
| G+Q+Nx, G+Q-Nx | 1.2G + 1.2Q +/- 1Nx |
| G+Q+Ny, G+Q-Ny | 1.2G + 1.2Q +/- 1Ny |

## LOAD CASES

- G: dead loads. Q: live loads.
- Gp1, Gp2: pattern dead loads. Qp1, Qp2: pattern live loads.
- Nx, Ny: notional loads in X and Y.

*Loading code: BS 6399. Concrete design code: BS 8110-1997.*

PERFORMED

# Beam and column design checks

## BEAMS: HIGHEST UTILISATION RATIOS IN THE DESIGN TABLE EXTRACT

| Axis | Storey | Beams | Utilisation | Status |
|---|---|---|---|---|
| A8 | 1 | 1B28, 1B27, 1B26, 1B25, 1B45 | 0.99 | Pass |
| A4 | 1 | 1B6, 1B7 | 0.93 | Pass |
| A10 | 1 | 1B11, 1B44 | 0.91 | Pass |
| A6 | 1 | 1B35, 1B36, 1B50 | 0.88 | Pass |
| A1 | 2 | 2B6, 2B2 | 0.87 | Pass |
| A15 | 1 | 1B13, 1B12 | 0.86 | Pass |
| A3 | 1 | 1B31, 1B32, 1B33, 1B46 | 0.86 | Pass |
| A2 | 1 | 1B23, 1B22, 1B43 | 0.81 | Pass |

## COLUMNS: DESIGN SUMMARY

| Item | Result |
|---|---|
| Section | 230 x 230 mm |
| Typical | 4 T16 (1.52 %), T10 links at 175 mm |
| Heavier columns | 4 T20 (2.38 %), T10 links at 225 mm |
| Highest utilisation | 0.94 (1C10) |
| Status | All rows in the extract pass |

- Every beam and column in the design table extracts shows a passing design status.
- In the extracts, the most heavily utilised beam is 0.99 and the most heavily utilised column is 0.94, both below 1.0.
- Utilisation ratio is design effect divided by section capacity, so lower is more reserve.

PERFORMED

# Slab design: first floor

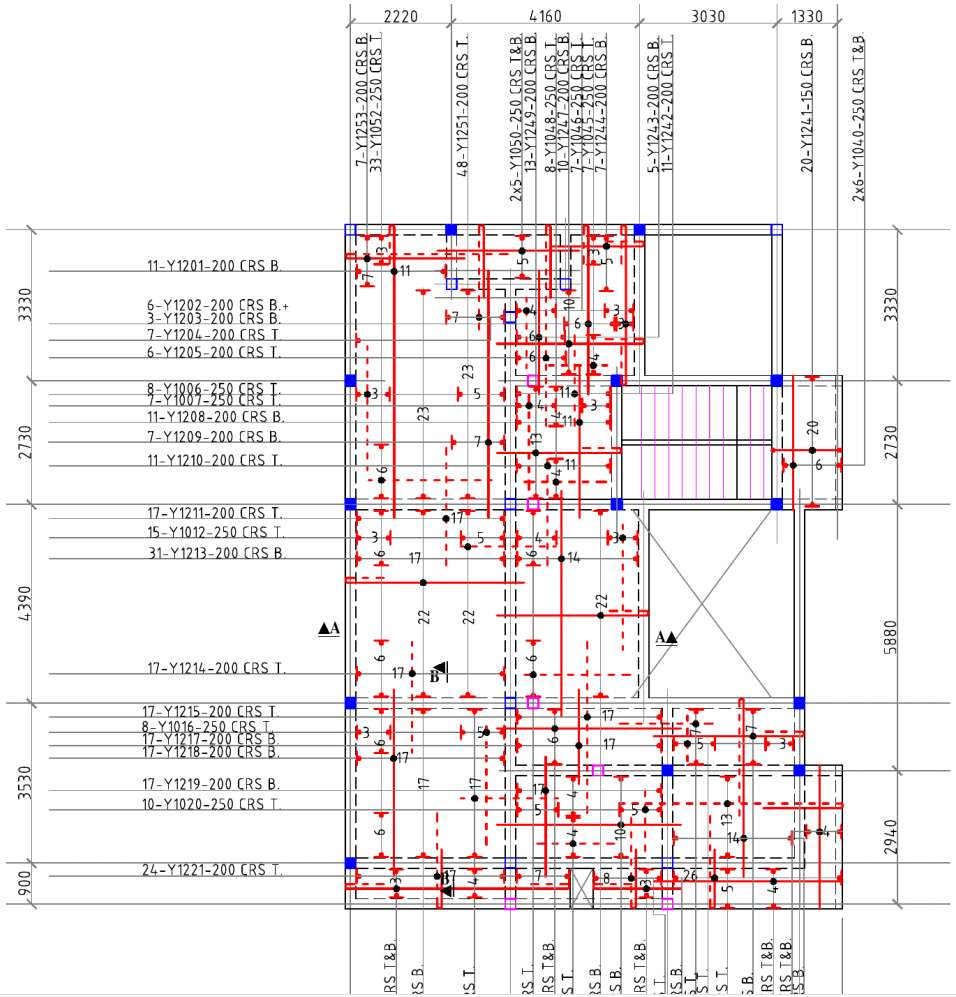



Figure 10. First floor slab reinforcement layout, panels P1 to P13.

## DESIGN SUMMARY

- Slab thickness: 150 mm, spanning between 230 mm wide beams.
- Designed to BS 8110-1997 with loading to BS 6399 (1.4G + 1.6Q).
- 13 slab panels, P1 to P13, each designed and detailed.
- Reinforcement: Y10 and Y12 bars, top and bottom, at 150 to 250 mm centres.
- Every bar group is marked with number, size, bar mark, spacing and position (T or B).
- Overall and panel dimensions are given on the layout.

PERFORMED

# Slab design: second floor and ground slab

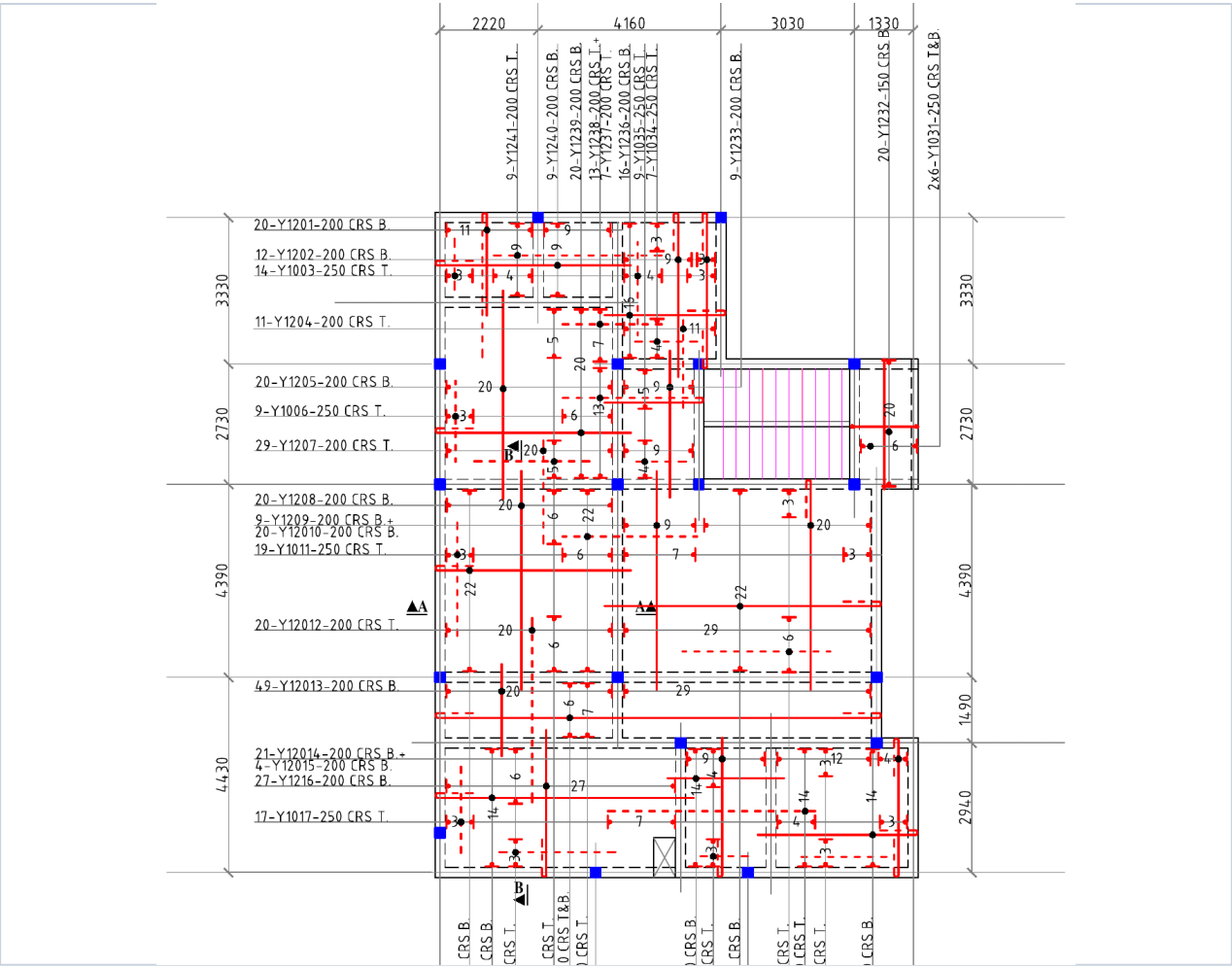



Figure 11. Second floor slab reinforcement layout, panels P1 to P12.

## DESIGN SUMMARY

- 150 mm slab, panels P1 to P12, designed to BS 8110-1997.
- Y10 and Y12 reinforcement at 150 to 250 mm centres, top and bottom, detailed on the layout.
- The layout shows the beams, the staircase opening and dimensions to the grid.
- Ground floor slab: 150 mm thick on fill, reinforced with A142 wire mesh.

PERFORMED

# Slab sections and reinforcement

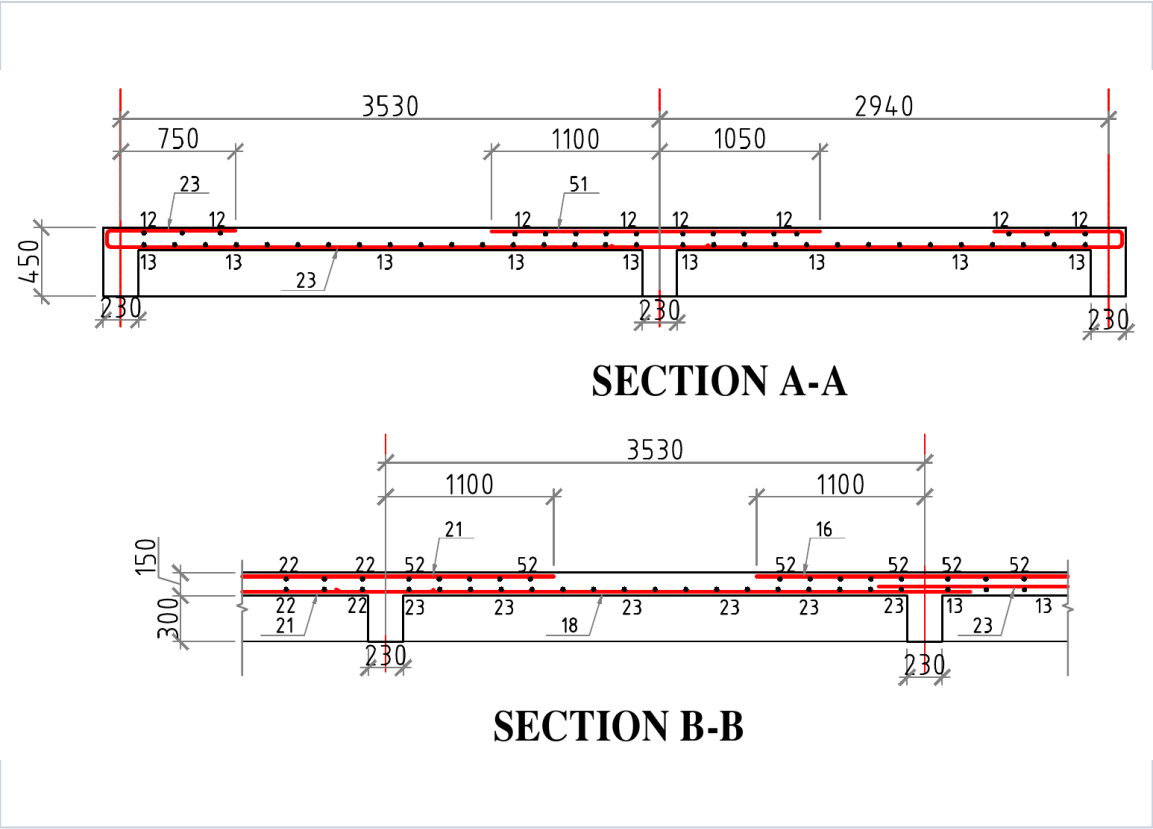



Figure 12. First floor sections A-A and B-B through slab and beams, with reinforcement.

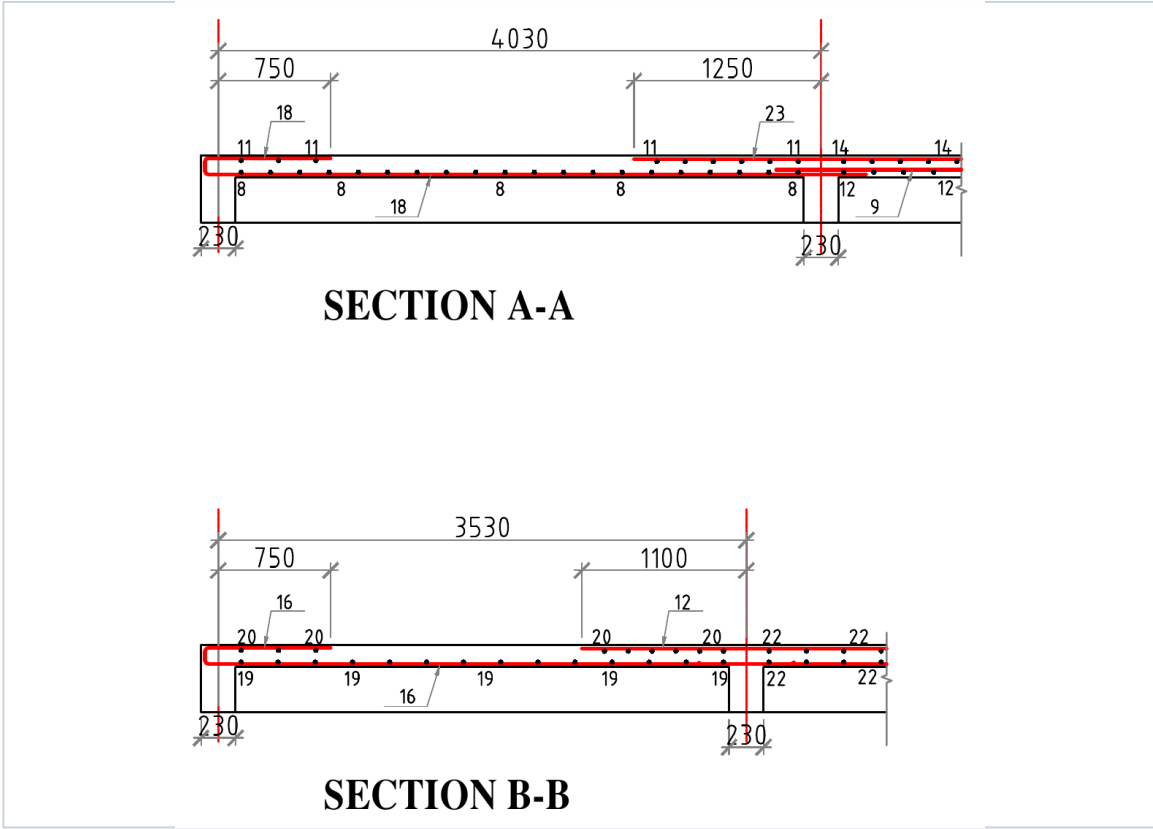



Figure 13. Second floor sections A-A and B-B through slab and beams, with reinforcement.

The sections show the 150 mm slab on the 230 mm beams, with top and bottom bars, their marks and curtailment lengths.

PERFORMED

# Foundation design basis and layout

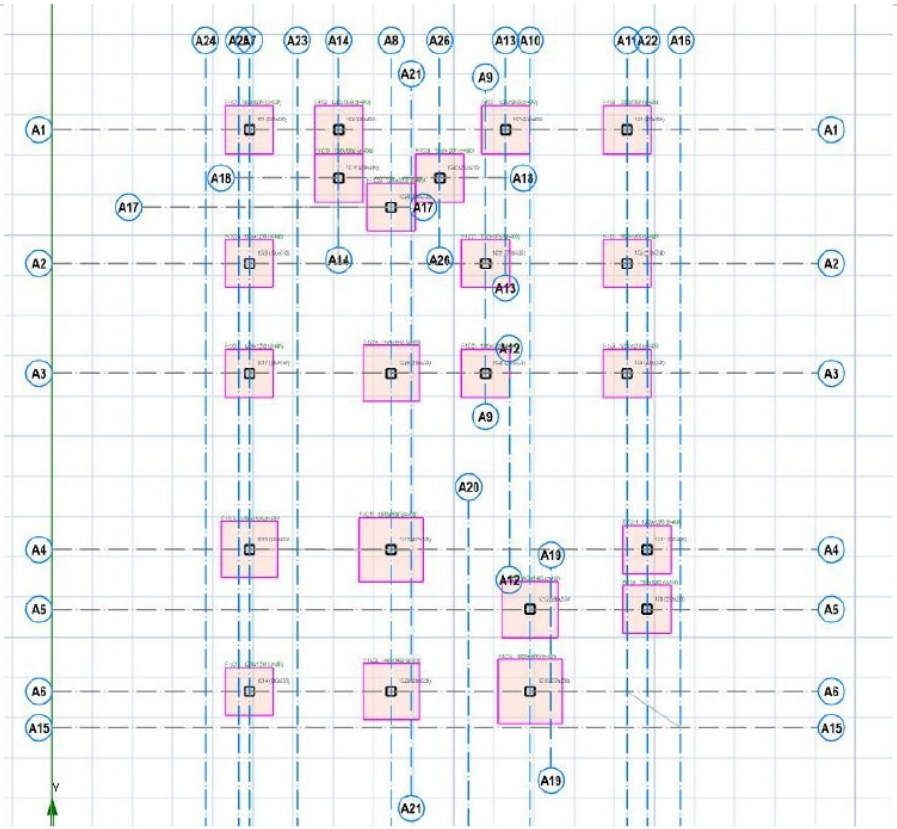



Figure 14. ProtaStructure foundation plan: isolated pad footing under every column.

## FOUNDATION DESIGN PARAMETERS IN THE MODEL

| | |
|---|---|
| Assumed allowable soil pressure | 200 kN/m2 |
| Soil unit weight | 18 kN/m3 |
| Ultimate strength factor on soil stress | 1.4 |
| Minimum pad base depth | 400 mm |
| Pad base concrete cover | 40 mm |
| Minimum steel ratio | 0.2 % |
| Bar size and maximum spacing | T16, 250 mm |

- Isolated pad footings are provided under every column.
- Footing marks B1 to B5 are shown on the foundation layout drawing.

PERFORMED

# Foundation layout and sections

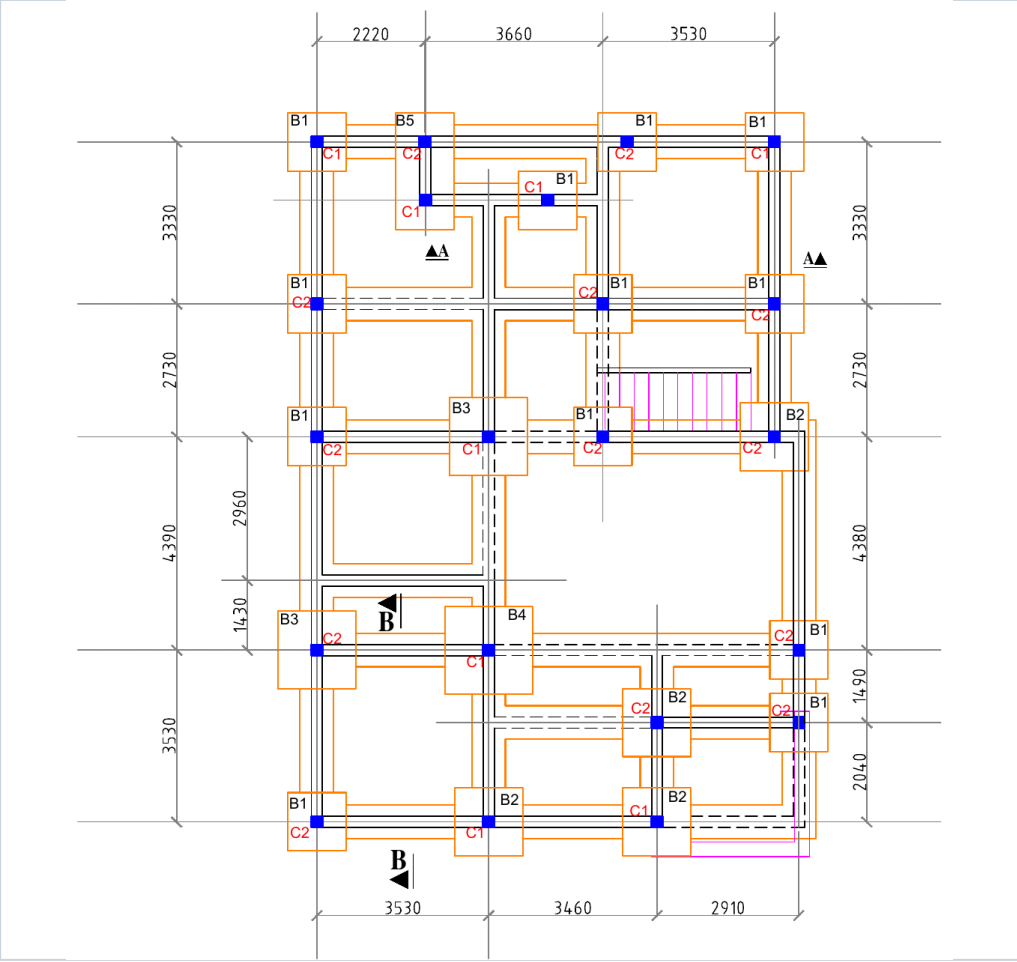



Figure 15. Foundation layout: footing marks B1 to B5 under column marks C1 and C2.

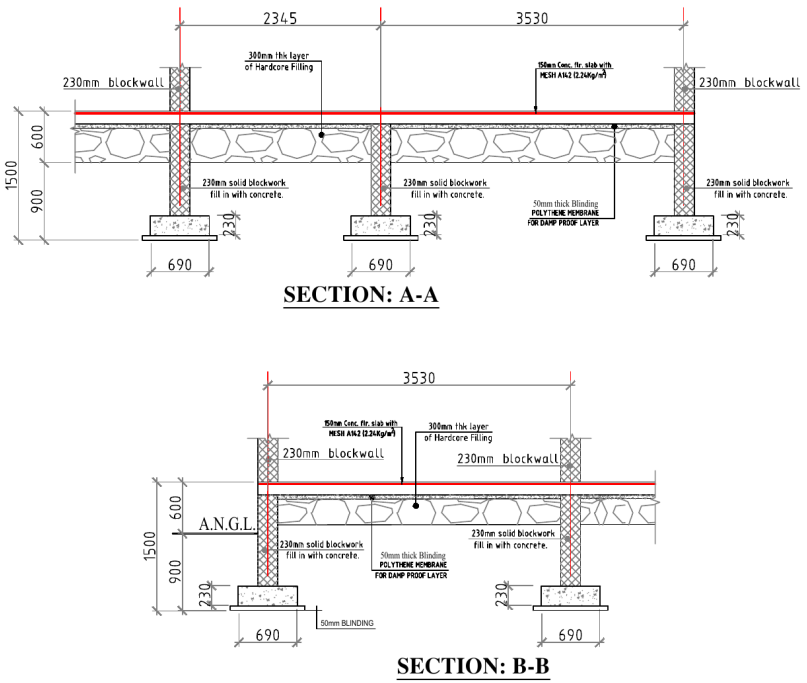



Figure 16. Sections A-A and B-B through the foundation, with 50 mm blinding.

- Ground floor slab: 150 mm thick, reinforced with A142 wire mesh.
- Blinding: 50 mm thick under footings.

PERFORMED

# Column base details

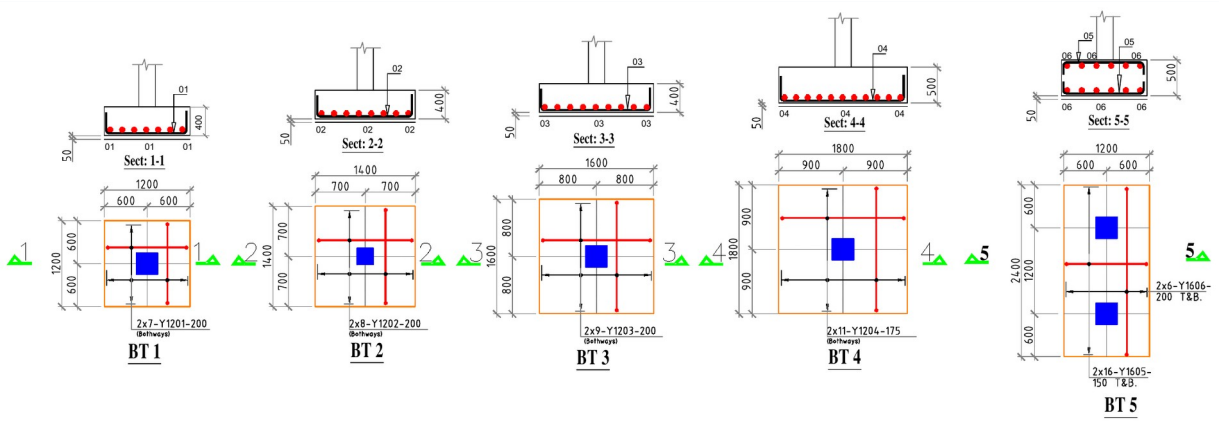



Figure 17. Column base details BT1 to BT5.

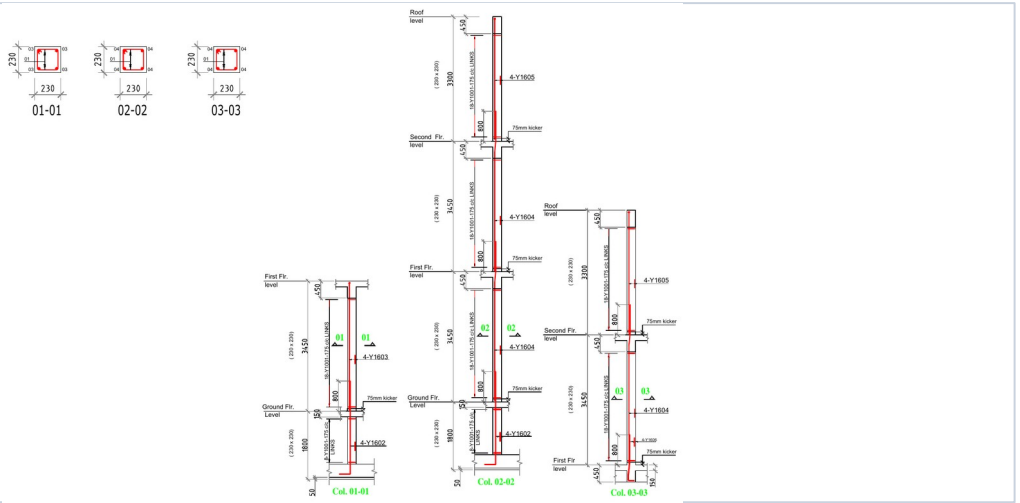
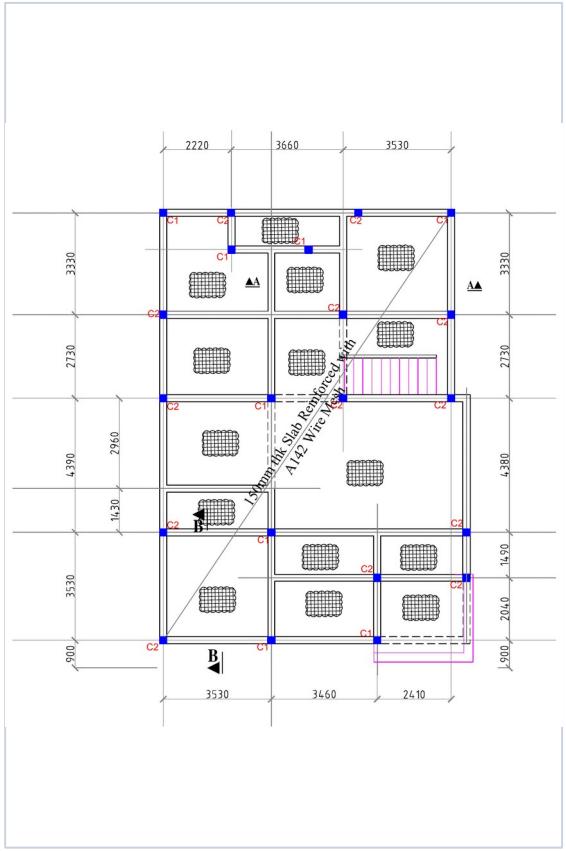



Figure 18. Column details by level.

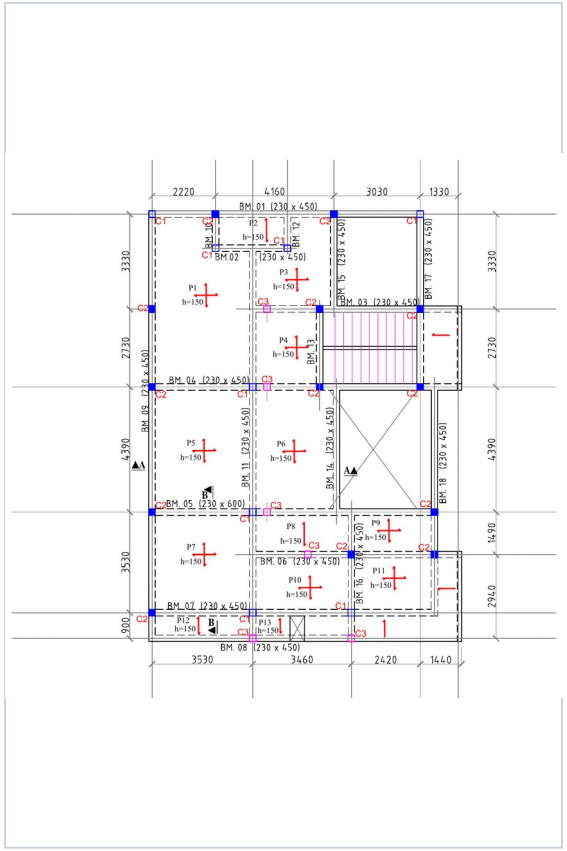
- Columns are 230 x 230 mm with 4 Y16 bars and Y10 links at 175 mm.
- 75 mm kickers shown at each level.

PERFORMED

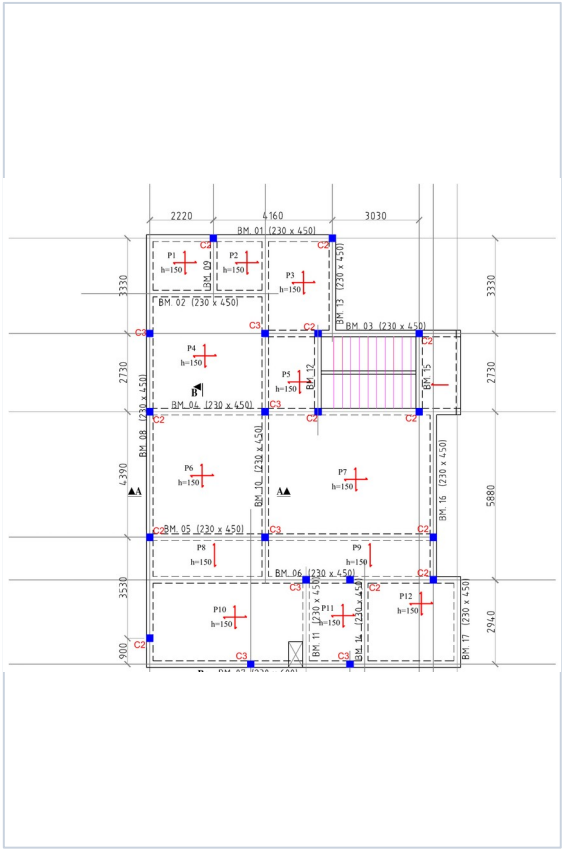
# Structural layouts by level



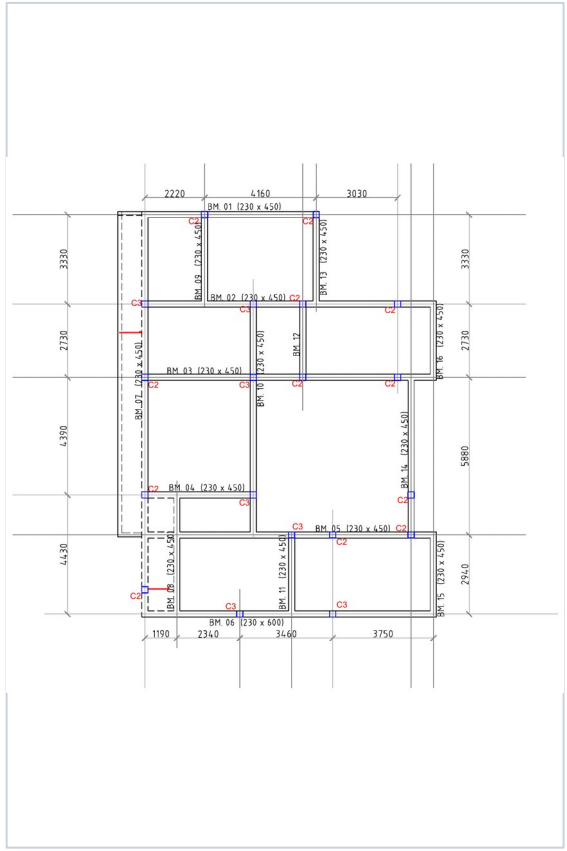

Ground floor layout



First floor layout



Second floor layout



Roof layout

Each layout dimensions the grid, columns and beams, and carries the section markers used on the detail sheets.

PERFORMED

# Beam reinforcement details

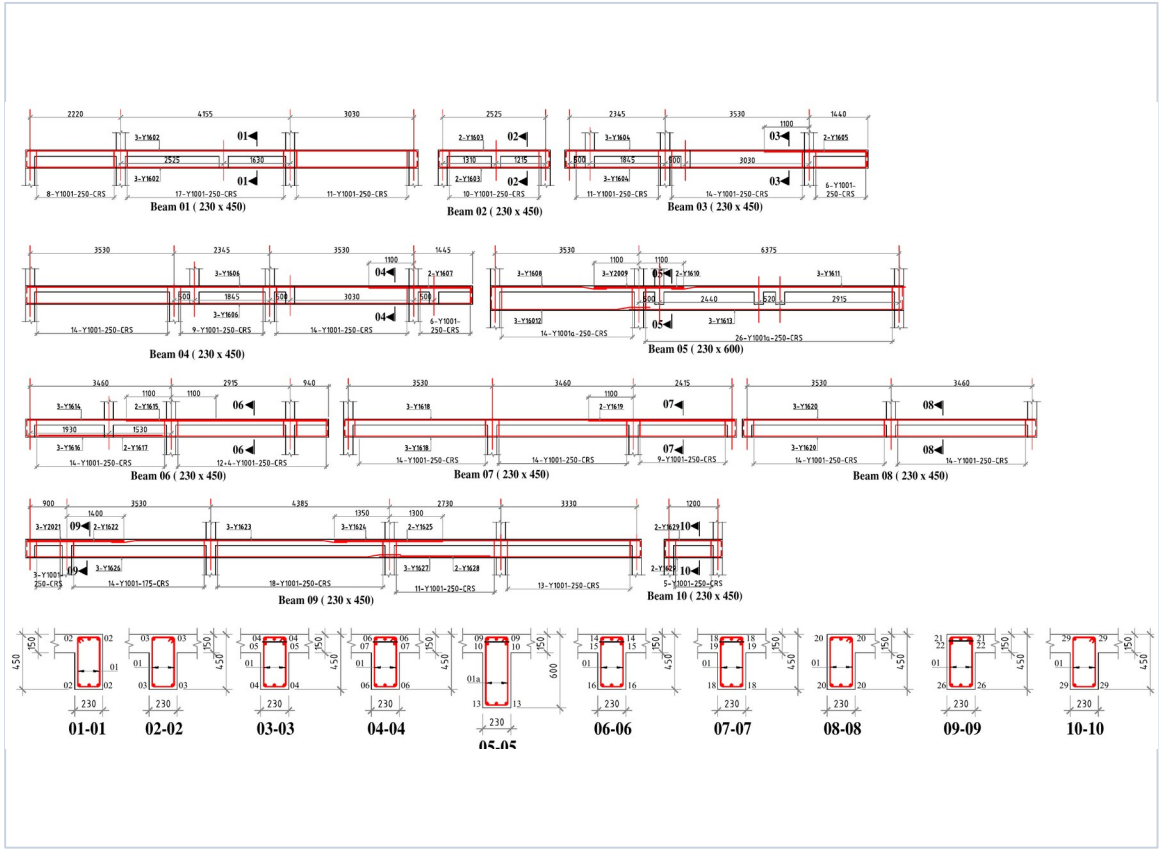



Figure 19. First floor beam reinforcement schedule and sections.

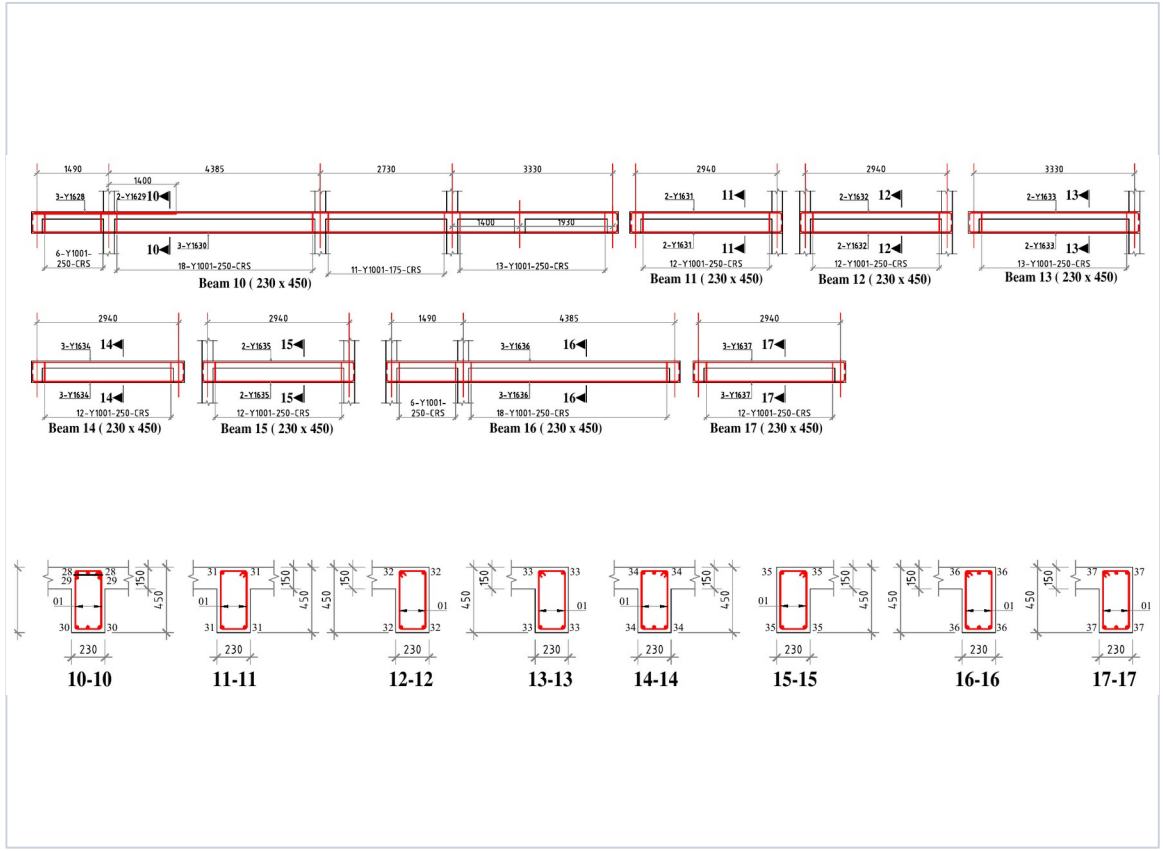



Figure 20. Second floor beam reinforcement details.

Beams are 230 x 450 mm unless marked 230 x 600 mm. Each beam shows span, top and bottom bars, link spacing and a numbered section.

PERFORMED

# Staircase details

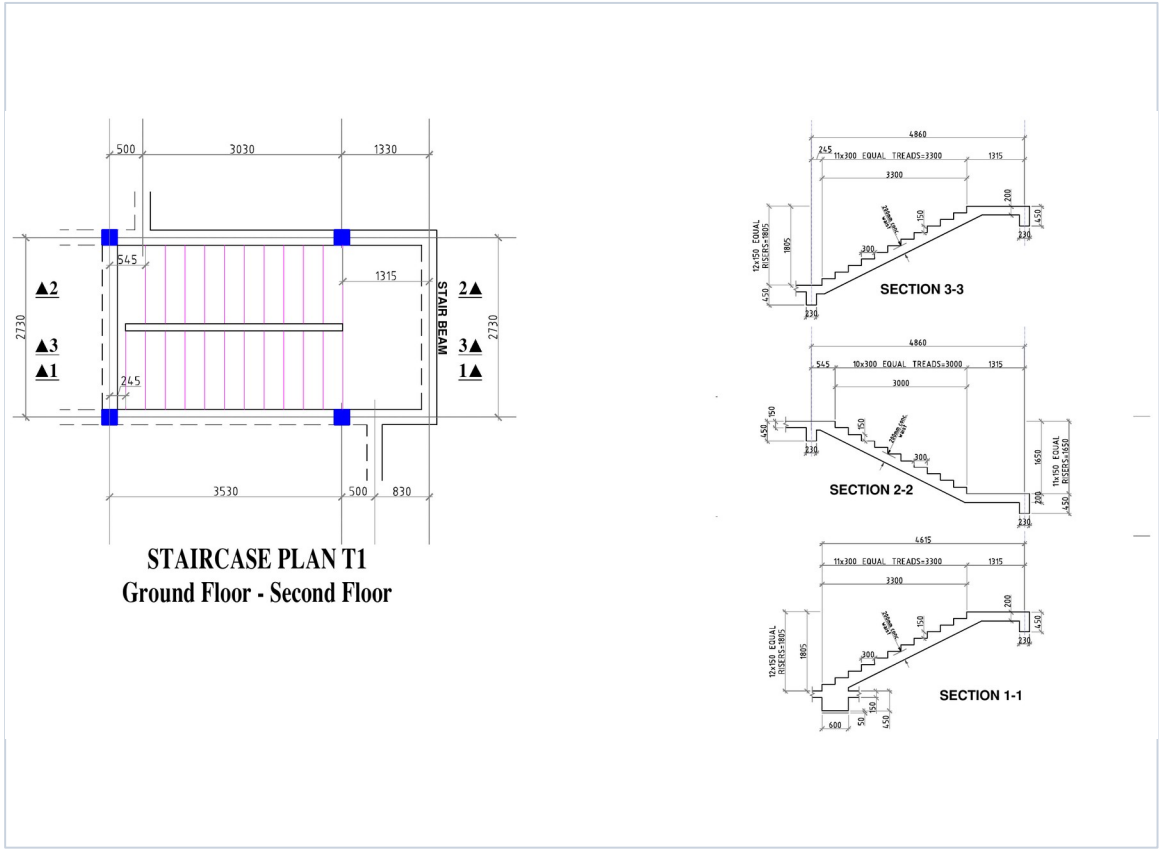



Figure 21. Staircase plan T1, ground floor to second floor, with sections 1-1, 2-2, 3-3.

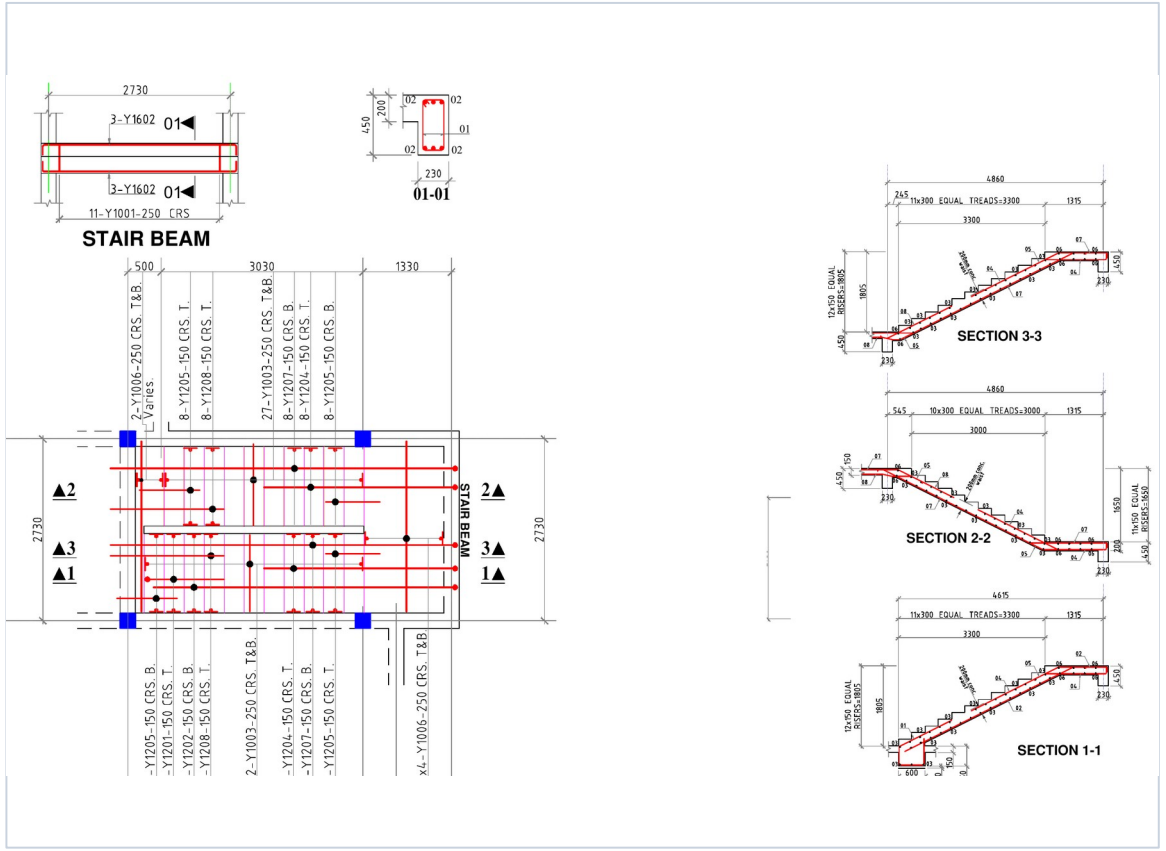



Figure 22. Staircase reinforcement and stair beam.

# Documentation package

| Package | Contents |
|---|---|
| Substructure | Foundation layout, sections A-A and B-B, column base details BT1 to BT5 |
| Framing layouts | Ground, first, second and roof layouts |
| Slab reinforcement | First and second floor slab reinforcement layouts with sections |
| Beams | First floor, second floor and roof beam reinforcement schedules and sections |
| Columns | Column details by level |
| Staircase | Plan T1, sections 1-1 to 3-3, reinforcement and stair beam |
| Total | 22 AutoCAD sheets, checked before issue |

# Contact


## Fahdil Dagold

STRUCTURAL DESIGN ENGINEER

Email fahdildagold07@gmail.com

LinkedIn linkedin.com/in/fahdil-dagold-029435222